EXERCISE – 6.2

SIMPLE LINEAR REGRESSION

1. The grades of a class of 9 students on a midterm report (*x*) and on the final examination (*y*) are as follows:

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

(a) Fit a linear regression model using the above data set also interpret the results.
(b) Estimate the final examination grade of a student who received a grade of 85 on the midterm report.
(c) Check the significance of slope of the SLRM.
(d) Check the significance of Intercept of the SLRM.
$(Ans: (a)\ \hat{y} = 12.06 + 0.78x + 12.16x_2 \quad (b)\ \hat{y} = 78\ )$

2. A study was made by a retail merchant to determine the relation between weekly advertising expenditures and sales.

Advertising Costs ($)	Sales ($)
40	385
20	400
25	395
20	365
30	475
50	440
40	490
20	420
50	560
40	525
25	480
50	510

(a) Fit a linear regression model using the above data set also interpret the results.
(b) Estimate the weekly sales when advertising costs are $35.
(c) Check the significance of slope of the SLRM.
(d) Check the significance of Intercept of the SLRM.
$(Ans: (a)\ \hat{y} = 343.71 + 3.22x \quad (b)\ \hat{y} = \$456.43\ )$

3. A professor in the School of Business in a university polled a dozen colleagues about the number of professional meetings they attended in the past five years (*x*) and the number of papers they submitted to refereed journals (*y*) during the same period. The summary data are given as follows:

$$n = 12, \quad \bar{x} = 4, \quad \bar{y} = 12,$$
$$\sum_{i=1}^{n} x_i^2 = 232, \quad \sum_{i=1}^{n} x_i y_i = 318.$$

(a) Fit a linear regression model using the above data set also interpret the results.
(b) Comment on whether attending more professional meetings would result in publishing more papers.
(c) Check the significance of slope of the SLRM.
(d) Check the significance of Intercept of the SLRM.
$(Ans: (a)\ \hat{y} = 37.8 - 6.45x\ (b)\ It\ appears\ that\ attending\ professional\ meetings\ would\ not\ result\ in\ publishing\ more\ papers.)$

4. Raw material used in the production of a synthetic fiber is stored in a place that has no humidity control. Measurements of the relative humidity and the moisture content of samples of the raw material (both in percentages) on 12 days yielded the following results:

Humidity	*Moisture content*
46	12
53	14
37	11
42	13
34	10
29	8
60	17
44	12
41	10
48	15
33	9
40	13

(a) Fit a linear regression model using the above data set also interpret the results.
(b) Predict the moisture content when the relative humidity is 38 percent.
(c) Check the significance of slope of the SLRM.
(d) Check the significance of Intercept of the SLRM.
$(Ans: (a)\hat{y} = 0.49 + 0.27x (b)\ \hat{y} = 10.75)$

5. A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:

Temperature, x	**Converted Sugar, y**
1.0	8.1
1.1	7.8
1.2	8.5
1.3	9.8
1.4	9.5
1.5	8.9
1.6	8.6
1.7	10.2
1.8	9.3
1.9	9.2
2.0	10.5

(a) Fit a linear regression model using the above data set also interpret the results.
(b) Predict the mean amount of converted sugar produced when the coded temperature is 1.75.
(c) Check the significance of slope of the SLRM.
(d) Check the significance of Intercept of the SLRM.
$(Ans: (a)\hat{y} = 6.41 + 1.81x\ (b)\ \hat{y} = 9.580\ \ (c)\ R^2 = \quad)$